VIRTA: A Deployable Embodied Virtual Trainer Using Smartphone Pose Sensing and Inertial Motion References for Exercise Feedback

MINAS ASLANYAN[1]
[1]Discrete Mathematics Department, Institute for Informatics and Automation Problems, National Academy of Sciences of the Republic of Armenia, Yerevan 0014, Armenia (e-mail: minas.aslanyan@iiap.sci.am; ORCID: 0009-0008-0094-3742)
Corresponding author: Minas Aslanyan (e-mail: minas.aslanyan@iiap.sci.am).

ABSTRACT Home exercise guidance has to observe the trainee, not merely demonstrate to them, yet the instrumentation that measures human movement accurately is absent from a living room. This article presents VIRTA, a deployable embodied virtual trainer that separates measurement from demonstration in time. Offline, an instructor wearing an inertial capture suit records a library of 180 exercises; the same recordings both animate a humanoid trainer and define the templates against which trainees are assessed. Online, the trainee wears no sensing instrumentation: a single smartphone runs BlazePose on device in 42 ms per frame, matches exercise templates by dynamic time warping, segments repetitions with a hysteresis phase detector at constant amortised per-frame cost, and evaluates per-exercise joint-angle constraints. Violations are realised as short imperative utterances and spoken by a lip-synchronised avatar on a virtual-reality headset. No image or landmark data leaves the phone. On a 2500-frame test set the mean per-joint position error is 28.5 mm under controlled conditions, 36.8 mm under partial occlusion and 42.1 mm in low light. Across 150 sessions, exercise classification reaches 94.0 % accuracy over the two assessed movements, repetition segmentation reaches 94.7 % temporal-overlap accuracy on squats and 92.4 % on rehabilitation postures, and form-error detection reaches precision 0.91 and recall 0.89 on squats. In a four-week single-arm deployment, 20 amateur exercisers reduced system-detected deviation events by 21.3 % and rated satisfaction 4.4 ± 0.5 out of five. The article reports the engineering detail needed to rebuild the system on commodity hardware.

INDEX TERMS Avatar animation, dynamic time warping, edge computing, exercise feedback, human pose estimation, inertial motion capture, mobile sensing, real-time systems, virtual reality.

I. INTRODUCTION

PARTICIPATION in home-based physical training has grown sharply, driven by on-demand video platforms and connected fitness hardware. Video-led training, however, is an open-loop medium: it demonstrates a movement from a viewpoint fixed by the producer, but it never observes the learner and therefore never corrects. This matters because in resistance and high-intensity training the difference between a productive repetition and an injurious one is frequently a joint-angle deviation of a few tens of degrees that the trainee cannot perceive without an external reference [1], [2]. Closing that loop couples a sensing problem with an interaction one: the system must measure the learner's kinematics, compare them against a trustworthy reference, and present the discrepancy through a channel the learner can follow while moving—here, an embodied virtual trainer that demonstrates the target movement on a virtual-reality headset and voices the correction with lip-synchronised speech—within a latency budget short enough to be corrective rather than retrospective.

Laboratory instrumentation solves the measurement problem but not the deployment problem. Optical motion capture delivers millimetre accuracy inside a calibrated volume, and body-worn inertial arrays deliver a few degrees of joint-angle accuracy [3], [4], but neither is present in a living room. Conversely, the sensor that is present—the smartphone camera—is monocular, uncalibrated and arbitrarily placed, and the inference must run within a mobile power and thermal envelope.

VIRTA resolves this tension by splitting the measurement across two tiers that are used at different times. A wearable inertial array is used offline to build a reference library: an instructor performs each exercise while wearing the suit, and the reconstructed kinematics serve both as the animation source for the virtual trainer and as the biomechanical standard against which trainees are later judged. A smartphone camera is used online as the only sensor present during a session. The reference tier can therefore be accurate without being portable, and the online tier can be portable without having to establish ground truth on its own.

A. CONTRIBUTIONS

The contributions of this article are practical rather than algorithmic. We do not introduce a new pose backbone, a new alignment algorithm or a new avatar representation; we show that established components can be composed into a closed-loop exercise trainer that runs entirely on hardware a household already owns, and we report the detail needed to rebuild it.

1) **A deployable sensing configuration.** During a session the trainee wears no sensing instrumentation and holds no tracked device, needs no depth sensor and no external tracker, and needs no second camera; the headset presents the trainer and carries no part of the measurement. All pose inference runs on the phone; neither images nor landmark coordinates leave the device (Sections III and V).
2) **A reference-driven architecture in which one captured motion serves two purposes.** The same inertially captured clip animates the visible exemplar and supplies the analysis template, so the demonstrated target is traceable to the assessment standard rather than authored separately (Section IV).
3) **An interpretable, auditable correction path.** Repetitions are segmented at constant amortised per-frame cost, per-exercise geometric constraints are evaluated over the completed repetition, and each violation maps to one of a finite inventory of authored utterances that names a body segment, a direction and a magnitude (Section V-D).
4) **Measured performance of the deployed configuration and the engineering detail required to reproduce it**: pose accuracy across three environmental conditions, repetition segmentation and error detection over 150 sessions, stage-level timing, and a four-week deployment (Sections VI and VII).

B. PRACTICAL APPLICATIONS

VIRTA is intended for deployment by three kinds of operator, at three cost points. A home user needs a smartphone and a consumer VR headset, already the two most common pieces of hardware in a connected-fitness household; the marginal hardware cost of adding form feedback to an existing home setup is zero, and the recurring cost is the cloud speech synthesis of a few short utterances per session. A gym or studio deploys the same configuration on a shared phone and headset, and gains the ability to record its own house technique into the reference library rather than accepting a vendor's; the capital cost is one inertial suit, which is an order of magnitude below an optical capture volume [3], [4], and it is paid once, offline, by the instructor rather than by every trainee. A rehabilitation provider uses the kinesiotherapy portion of the library for supervised home exercise between clinic visits, where the constraint that the patient wear no sensors is not a convenience but a precondition for adherence.

Three deployment constraints shape the design throughout and are worth stating at the outset. First, the trainee is under exertion and cannot read a display, so corrections must be spoken and rate-limited. Second, a single camera facing the trainee cannot resolve depth reliably, so the assessment is restricted to quantities that survive that limitation—joint angles and torso-normalised distances—rather than absolute 3D positions. Third, camera imagery of a person exercising at home is sensitive, so the pipeline is built such that no frame and no landmark leaves the phone.

C. RELATION TO THE AUTHOR'S EARLIER WORK

The conference article [5] introduced the system. This article extends the architectural and algorithmic account, adds the deployment framing above, and reports in full the evaluation of the author's dissertation [6], Chapter 6, of which this system is the subject. Two related articles by the author address adjacent questions: a design and implementation study of mobile pose estimation and action recognition [7], and a comparative evaluation of inertial against vision-based capture [8]. The overlap between those sources and the results reported here is identified explicitly in Section VI.

II. RELATED WORK

A. WEARABLE INERTIAL SENSING FOR HUMAN KINEMATICS

Body-worn inertial measurement units estimate segment orientation by fusing accelerometer, gyroscope and magnetometer signals, and have become the standard portable alternative to optical capture. López-Nava and Muñoz-Meléndez [3] survey the field and identify the two characteristic error sources that also govern our reference tier: integration drift, which accumulates over a take and manifests as segment sliding, and magnetometer disturbance from ferrous structures and electrical equipment. Mihcin et al. [9] evaluate a commercial 19-sensor suit for biomechanical modelling, and Ciklacandir et al. [4] compare inertial against video-based capture on joint angles, reporting joint-level limitations relevant to the reference tier. The systematic review of Scataglini et al. [10] finds markerless camera systems good to excellent against marker-based references on spatiotemporal parameters while remaining weaker on fine joint kinematics—the same asymmetry that motivates our split between an instrumented reference tier and a camera-only online tier. These results support using an inertial array as a reference instrument provided its own accuracy is quantified rather than assumed.

B. MONOCULAR 3D POSE ESTIMATION ON EMBEDDED PLATFORMS

Recovering 3D pose from a single RGB stream is underdetermined and sensitive to body proportion, self-occlusion, viewpoint and background. Progress has been dataset-bound: 2D corpora such as LSP [11] and MPII [12] provide approximate joint annotations without segmentation, while 3D corpora such as HumanEva [13] and Human3.6M [14] provide accurate ground truth but were captured in a small number of laboratory settings, limiting generalisation to domestic environments. Architecturally, the dominant decomposition estimates 2D keypoints and lifts them to 3D; stacked hourglass networks [15] established the encoder–decoder heatmap formulation on which most subsequent work builds. For embedded operation, BlazePose [16] is the design we adopt: a lightweight detector derived from BlazeFace [17] runs only on tracking loss, and a tracker network predicts 33 landmarks, a presence flag, and the region of interest for the next frame. Heatmap and offset heads supervise an embedding during training and are removed before inference, leaving a regression network small enough for a mobile GPU. The design and implementation trade-offs of running this class of model inside a mobile application are treated separately [7].

C. EXERCISE SEGMENTATION AND REPETITION COUNTING

Counting repetitions is a prerequisite for per-repetition assessment. Accelerometer-based approaches exploit the quasi-periodicity of resistance exercise: Pernek et al. [18] detect repetitions by template matching with adaptive thresholds, and Morris et al. [19] segment and count from a single wearable. Dynamic time warping [20], [21] is the standard tool for comparing temporally misaligned executions of the same movement. Vision-based counting has since been posed as dense period prediction over video, at a computational operating point we do not adopt: because VIRTA already holds a reference clip for every exercise, a per-exercise template is available at no cost and a general-purpose period predictor is unnecessary.

D. VIRTUAL TRAINERS AND MOVEMENT GUIDANCE IN VR AND AR

How a correction is presented decides whether it can be acted on mid-movement. Chua et al. [22] established the paradigm of learning a physical skill from a virtual instructor, and the line continues in recent work: Tian et al. [23] drive a group-based Tai Chi trainer across multiple standalone headsets, displaying real-time action guidance trajectories and comparing traditional coach guidance against VR and mixed reality. PoseCoach [24] is the closest analogue to our feedback stage, comparing a novice's running pose against an expert's over user-definable pose attributes and rendering the difference on a 3D human model rather than reporting a score. In AR, Wu et al. [25] overlay visual cues on at-home workout videos and show that the cues let users adjust their movements from real-time feedback. Yu et al. [26] organise this space by separating feedforward cues from corrective feedback—a split VIRTA realises concretely, the avatar supplying the demonstration and the constraint bank of Section V-E the correction. Eom et al. [27] report an embodied exergame evaluated with ICU patients, a clinical population and intervention distinct from the present exercise setting. Egocentric methods such as EgoPoseVR [28] instead recover full-body pose from headset-centred inputs.

E. AUTOMATIC ASSESSMENT OF MOVEMENT QUALITY

Hülsmann et al. [2] classify motor errors in squats and Tai Chi pushes and deliver real-time feedback as predefined verbal information and as colour highlights on the trainee's avatar, establishing that error-specific feedback is actionable. Fieraru et al. [1] reconstruct 3D pose, shape and motion, segment repetitions, and produce localised quantitative feedback against standards learnt from trainers. Both require instrumentation or viewpoints that a domestic deployment does not have. Rule-based assessment, in which explicit geometric constraints replace a learned classifier, trades coverage for interpretability and for independence from labelled training data; Stenum et al. [29] demonstrate that pose-estimation output supports clinically meaningful kinematic measurement when normalised appropriately. VIRTA adopts the rule-based route for the reasons set out in Section V-E.

F. POSITIONING

Table 1 places VIRTA against the five closest systems on the five axes that determine whether a movement-guidance system can be deployed outside a laboratory: what senses the trainee, what the trainee has to wear or install, how specific the resulting feedback is, through which channel it arrives, and where the system has been shown to work. Entries are taken from the cited articles; cells that those articles do not state are marked rather than inferred.

Read down the instrumentation column, the distinguishing property of VIRTA is that the column is empty and the loop closes during the session. PoseCoach asks nothing of the user but analyses recorded video offline; AIFit asks nothing body-worn, but its stronger multi-view mode presumes four calibrated cameras, which a living room does not have. The remaining systems each require a capture installation, a headset that is also the sensor, or an AR device. VIRTA keeps sensing on a commodity smartphone and puts only presentation on the headset. The cost of that choice is the single-viewpoint limitation examined in Section IX.

III. SYSTEM ARCHITECTURE

A. SENSING TIERS AND REQUIREMENTS

VIRTA comprises an offline reference tier and an online measurement tier, and at runtime it splits across two devices: a smartphone that performs all sensing and analysis, and

TABLE 1. VIRTA Against the Closest Movement-Guidance Systems. Entries Are Sourced From the Cited Articles; Unstated Details Are Marked.

System	Sensing modality	Instrumentation required of the user	Feedback granularity	Feedback channel	Deployment setting
VIRTA (this work)	Monocular RGB, single smartphone camera; on-device BlazePose	None. Phone on a stand; the inertial suit is worn only by the instructor, offline	Per repetition, per joint angle: names segment, direction and magnitude	Spoken correction from a lip-synchronised avatar in VR	Home; commodity phone and consumer headset
Fieraru *et al.* [1] (AIFit)	RGB video; 3D pose, shape and motion reconstruction	None body-worn. Evaluated single-view and multi-view; the multi-view mode uses four synchronised RGB cameras with known parameters	Per repetition; localised quantitative deviation from trainer-learnt standards	Human-interpretable text and visual feedback	Stated by the authors as usable at home, outdoors or at the gym
Hülsmann *et al.* [2]	Marker-based optical capture in a VR coaching installation: OptiTrack, ten Prime 13W cameras, 19 joints at 120 Hz [30]	A customised motion-capture suit carrying 41 passive markers, with the arm and hand markers taped to the skin [30]	Per error class, from a trained classifier over selected features	Predefined verbal information plus colour highlights on the trainee's avatar	Laboratory VR coaching installation
Tian *et al.* [23]	Multiple standalone VR devices; body-tracking detail not stated	VR or MR headset; controller or tracker detail not stated	Whole-body: real-time action guidance trajectories	Visual guidance trajectories in VR and MR, with avatar companions	Multi-user Tai Chi training; VR and MR compared against coach guidance
Liu *et al.* [24]	Monocular running video	None; the user supplies a video	User-definable pose attributes, e.g. joint angles and step distances	Part-based 3D animation on a human model, in a visual-analytics interface	Offline desktop analysis of recorded running video
Wu *et al.* [25]	Pose tracking; sensing device not stated	AR display; display and tracker detail not stated	Real-time cues supporting in-session movement adjustment	AR visual cues overlaid on the workout video	At-home following of workout videos, in AR

a virtual-reality headset that renders the embodied trainer and delivers the spoken, lip-synchronised feedback, the two communicating over WebRTC on the local network (Section V-E3). The headset is a presentation device only: it performs no sensing, and the trainee holds no controllers during an exercise set. The design requirements follow from the deployment target set out in Section I-B.

- **R1** The online sensor set must not exceed one smartphone. The trainee must wear no sensing instrumentation and hold no tracked device; the headset is for presentation only.
- **R2** Visual inference must sustain $\geq$20 fps within a mobile thermal envelope.
- **R3** Reported deviations must name a body segment, a direction and a magnitude, since a scalar quality score is not actionable.
- **R4** Image data must not leave the device.
- **R5** Feedback must be rate-limited so that a trainee under exertion receives at most one instruction per repetition.

Fig. 1 shows the resulting architecture and Fig. 2 isolates the online measurement chain. Camera frames are delivered to the inference stage as GPU buffers, avoiding a per-frame host copy; landmark coordinates are consumed in process and discarded, so the only data crossing a network boundary is the short feedback token described in Section V-E3. In the diagrams, "frontal camera" denotes a camera placed in front of and facing the trainee; the evaluated handset used the main rear camera specified in Table 2.

B. RECOMMENDED DEPLOYMENT GEOMETRY

Because the camera is placed by the user rather than by an installer, the system ships with a recommended room geometry, illustrated in Fig. 3. The requirement the geometry has to satisfy is that the whole body remains in frame throughout the movement, including at the bottom of a squat and in floor-based postures; the application enforces this at session start by refusing to proceed until all 33 landmarks are present.

C. LATENCY BUDGET

The delay Δ between the frame containing a deviation and the first audible phoneme of the corresponding correction decomposes as

$$\Delta = \Delta_{\mathrm{cap}} + \Delta_{\mathrm{inf}} + \Delta_{\mathrm{filt}} + \Delta_{\mathrm{seg}} + \Delta_{\mathrm{eval}} + \Delta_{\mathrm{net}} + \Delta_{\mathrm{tts}}, \quad (1)$$

covering capture, inference, filtering, segmentation, constraint evaluation, network transfer and speech synthesis. The terms are not comparable in magnitude. Six are ordinary system latency terms, and Section VII-D reports the measured ones. The seventh, Δ_{seg}, is structural: a statement about a repetition cannot be emitted before that repetition has ended, so for a squat performed at approximately 2 s per repetition this term alone reaches 2 s. VIRTA therefore targets

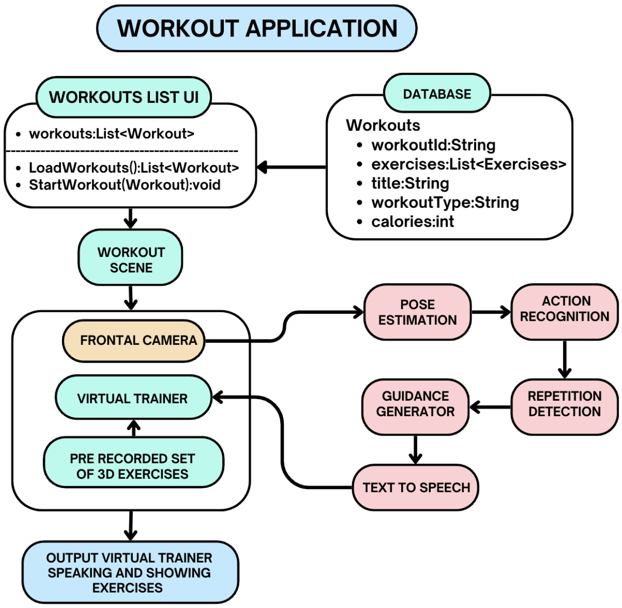

FIGURE 1. System architecture. The presentation path loads a session from the exercise database and plays the retargeted instructor animation on the virtual trainer. The measurement path consumes the camera stream and performs pose estimation, exercise classification, repetition detection and guidance generation; the resulting text is synthesised to speech and spoken by the trainer with phoneme-synchronised visemes.

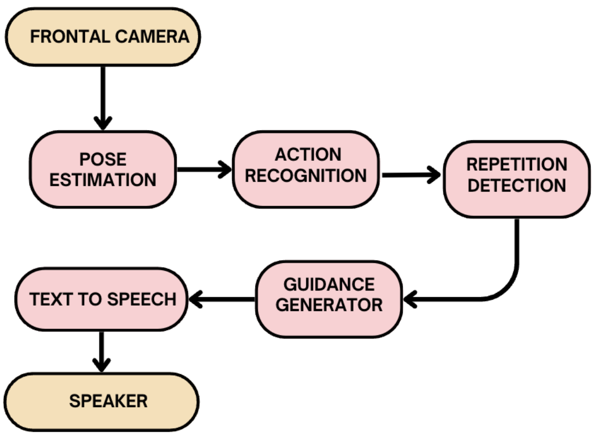

FIGURE 2. Online measurement chain: camera → 3D pose estimation (Section V) → exercise classification and repetition segmentation (Section V-D) → constraint evaluation and guidance generation (Section V-E) → speech synthesis.

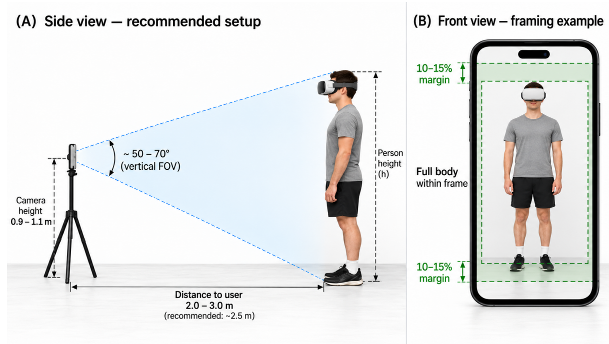

FIGURE 3. Recommended deployment geometry for a home session. Distances, camera height, field of view and margins are setup guidance issued to the user, not measured specifications of the apparatus used in the evaluation of Section VII.

correction of the next repetition rather than the current one, and R5 exists so that an instruction referring to a completed repetition does not arrive in the middle of the following one.

D. NOTATION

Let $J = 33$ be the number of landmarks, $\mathbf{p}_j^{(t)} \in \mathbb{R}^3$ the position of landmark j at frame t, $\mathbf{P}^{(t)}$ the corresponding pose, and $c_j^{(t)} \in [0, 1]$ the model's visibility score. Let $\mathcal{A}$ denote the exercise repertoire and $a \in \mathcal{A}$ the exercise being performed.

IV. REFERENCE CAPTURE TIER

A. RATIONALE

Animating the virtual trainer requires a trajectory of 84 bone transforms over time. In principle the monocular pipeline of Section V could supply it, requiring no additional hardware. We reject this because the reference sequence is not merely displayed: it also defines the standard against which every trainee is judged, so reconstruction error in the reference propagates as systematic bias into every subsequent assessment. The reference tier is therefore instrumented, and the monocular tier is used only for the trainee.

B. INSTRUMENTATION AND PROTOCOL

Capture uses a commercial suit, the Rokoko Smartsuit Pro II, carrying nine-degree-of-freedom inertial measurement units [9] (tri-axis accelerometer, gyroscope and magnetometer each) streaming at 100 Hz (Fig. 4). All 19 IMU

Number	Sensor Location	Segment Label	Joint Label
1	head	Head	Neck
2	right pelvis	Pelvis	Pelvis
3	right upper leg	RightUpperLeg	RightHip
4	right lower leg	RightLowerLeg	RightKnee
5	right foot	RightFoot	RightAnkle
6	left pelvis	Pelvis	Pelvis
7	left upper leg	LeftUpperLeg	LeftHip
8	left lower leg	LeftLowerLeg	LeftKnee
9	left foot	LeftFoot	LeftAnkle
10	right chest	Chest	Thorax
11	right shoulder	RightShoulder	RightScapula
12	right upper arm	RightUpperArm	RightShoulder
13	right fore arm	RigthForeArm	RightElbow
14	right hand	RightHand	RightWrist
15	left chest	Chest	Thorax
16	left shoulder	LeftShoulder	LeftScapula
17	left upper arm	LeftUpperArm	LeftShoulder
18	left fore arm	LeftForeArm	LeftElbow
19	left hand	LeftHand	LeftWrist

FIGURE 4. Sensor placement, with the body segment and joint each unit drives.

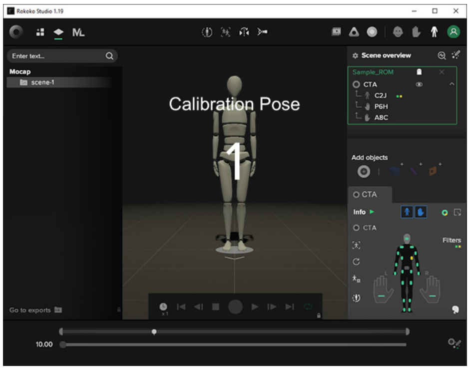

FIGURE 5. T-pose calibration performed before each capture session.

sensors were active during reference-motion capture. Sensors are cabled to a wireless hub powered by a 5000 mA h battery, giving approximately 6 h of continuous operation. Each session begins with a neutral-pose calibration (Fig. 5) that establishes the sensor-to-segment alignment; capture then proceeds without further intervention. Live monitoring, trimming and jitter filtering are performed in Rokoko Studio. The approach is portable and inexpensive relative to optical systems [3], [4], which is what makes recording a library of this size feasible outside a laboratory.

C. ERROR SOURCES AND POST-PROCESSING

Inertial capture exhibits two characteristic failure modes. Finite sensor precision produces drift that accumulates over a take and manifests as skeletal sliding, and the magnetometer component is sensitive to ambient magnetic disturbance [3]. The capture software flags magnetic interference during recording and provides trimming and smoothing filters. Residual root-translation error was corrected by direct editing of the animation F-curves (Fig. 6). Clips were exported in FBX and retargeted onto the humanoid avatar rig (Fig. 7), after which any clip in the library drives any avatar without per-clip authoring.

D. REFERENCE LIBRARY

The library comprises 120 high-intensity interval training exercises and 60 kinesiotherapy (musculoskeletal injury-relief) exercises, giving 180 distinct movements; the latter category

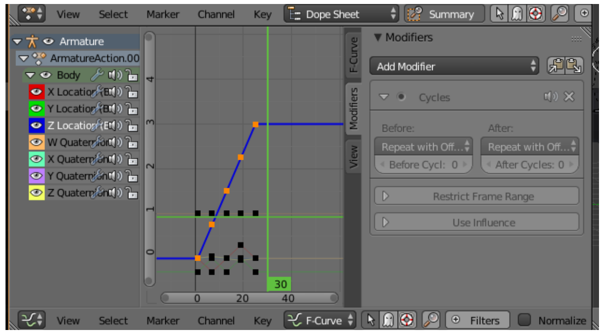

FIGURE 6. Residual drift correction: root translation and rotation F-curves are edited directly to remove accumulated positional error before export.

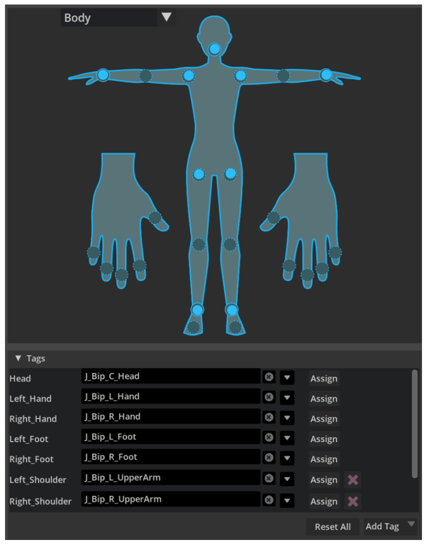

FIGURE 7. Retargeting the captured skeleton onto the humanoid rig. Each source bone is bound to the corresponding humanoid slot.

is what Section VII reports as the "rehabilitation" class. Each is stored as a retargeted animation clip and as the associated joint-angle trajectory, the latter serving as the template for classification (Section V-D1) and as the constraint reference (Section V-E).

Clips that retained visible artefacts after the automatic filters were repaired by hand as described in Section IV-C.

V. ONLINE SENSING AND ON-DEVICE PROCESSING

Once the reference library of Section IV is in place, the whole of the online session runs from a single smartphone camera. This section describes the on-device chain end to end: reconstructing a 3D body model from each frame (Section V-A), normalising and filtering it (Section V-B), identifying the exercise and segmenting its repetitions (Section V-D), and evaluating the geometric constraints whose violations become spoken corrections (Section V-E).

A. VISUAL POSE ESTIMATION

The measurement tier reconstructs a $J = 33$ landmark body model per frame using BlazePose [16] (Fig. 8). A lightweight detector adapted from BlazeFace [17] locates the subject and

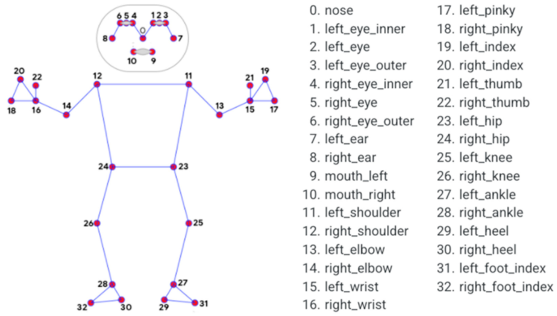

FIGURE 8. Landmark topology recovered from the camera facing the trainee.

FIGURE 9. Measurement tier running on the smartphone: the author is segmented and the 33 reconstructed landmarks overlaid on the camera preview.

predicts the mid-hip point used to align subsequent stages; a tracker then predicts landmark coordinates, a presence flag, and the refined region of interest for the next frame, so the detector runs only on tracking loss. The network combines heatmap, offset and regression heads; the heatmap and offset branches supervise the embedding during training and are removed before inference, following the encoder–decoder formulation of [15].

We deploy the 1.3 M-parameter variant. Its published training set comprises 85 000 hand-annotated images with simulated occlusion, of which 25 000 depict a single person performing fitness exercises—a distribution close to our deployment, which is a further reason for the choice. The model is quantised to 8-bit and pruned before deployment, and executes through the MediaPipe GPU delegate. Fig. 9 shows the pipeline running on the target device.

B. NORMALISATION

Raw landmark coordinates depend on camera placement, subject distance and subject size, none of which is controlled. Lengths are therefore expressed as ratios of torso length, defined as the mean distance from the neck landmark to the left and right hip landmarks [29]:

$$\ell_{\text{torso}} = \tfrac{1}{2}\left(\|\mathbf{p}_{\text{neck}} - \mathbf{p}_{\text{hip}}^{L}\| + \|\mathbf{p}_{\text{neck}} - \mathbf{p}_{\text{hip}}^{R}\|\right), \quad \tilde{d} = \frac{d}{\ell_{\text{torso}}}. \tag{2}$$

Torso length is close to constant across the frames of a session, which is what makes it a usable denominator: a normalised upper-arm length of 0.6 means the upper arm is 0.6 of the torso, for any subject at any distance. The angular features that carry most of the constraint bank (Section V-E) are scale-invariant by construction and require no normalisation.

C. TEMPORAL FILTERING AND CONFIDENCE

Per-frame landmark estimates carry high-frequency tracking noise, which the angular features differentiate. A trailing mean over a window of $N = 30$ frames is applied to the tracked coordinate within the segmentation stage (Eq. (5)). The model also returns a per-landmark visibility score $c_j^{(t)}$; the evaluated implementation records it but does not use it to gate constraint evaluation, a design choice revisited in Section IX.

D. ON-DEVICE SIGNAL PROCESSING

Per-repetition assessment requires knowing which exercise is being performed and where each repetition begins and ends. The two problems are solved differently because their duty cycles differ: classification runs once per set and can afford an $O(NM)$ alignment, whereas segmentation runs at frame rate alongside network inference.

1) Exercise Classification

Dynamic time warping [20], [21] compares two sequences that follow the same trajectory at different rates. For series $X = (x_1, \ldots, x_N)$ and $Y = (y_1, \ldots, y_M)$, a warping path $w_1, \ldots, w_K$ with $w_k = (a, b)$ linking x_a to y_b must satisfy boundary ($w_1 = (1, 1)$, $w_K = (N, M)$), monotonicity and continuity constraints. The accumulated cost follows the recursion

$$\begin{aligned} D(a,b) = d(x_a, y_b) + \min\{&D(a-1,b),\ D(a,b-1),\\ &D(a-1,b-1)\}, \end{aligned} \tag{3}$$

with $d(x_a, y_b) = \|x_a - y_b\|^2$, giving $\mathrm{DTW}(X,Y) = D(N,M)$ in $O(NM)$ time. The optimal path accommodates executions of the same movement at different tempos.

Classification is nearest-neighbour under this metric. The candidate sequence is compared against every reference template and the closest label returned:

$$\hat{a} = \arg\min_{a \in \mathcal{A}} \ \mathrm{DTW}(X, T_a), \tag{4}$$

where T_a is the template for exercise a, at cost $O(|\mathcal{A}|NM)$ per classification. The returned label selects the constraint set applied for the rest of the set, so its accuracy bounds everything downstream (Section IX).

Algorithm 1 Online repetition segmentation

Require: axis A; primary landmark L^{p}, optional secondary L^{s}; window N; dead band Δ_{thresh}

1: $B \leftarrow$ FIFO buffer of capacity N; $r \leftarrow 0$; $\mathrm{up}_{\mathrm{prev}} \leftarrow$ FALSE
2: **for** each arriving frame t **do**
3: $\quad p_t \leftarrow$ coordinate A of L^{p}_t; push p_t into B
4: $\quad \bar{p}_t \leftarrow \mathrm{mean}(B)$
5: $\quad$ compute $\mathrm{down}_t, \mathrm{up}_t$ by Eq. (6)
6: $\quad$ **if** L^{s} provided **then**
7: $\quad\quad$ repeat lines 3–5 for L^{s} and conjoin the flags
8: $\quad$ **end if**
9: $\quad$ **if** $\mathrm{down}_t \wedge \mathrm{up}_{\mathrm{prev}}$ **then**
10: $\quad\quad r \leftarrow r + 1$; **emit** repetition boundary at t
11: $\quad$ **end if**
12: $\quad \mathrm{up}_{\mathrm{prev}} \leftarrow \mathrm{up}_t$
13: **end for**

2) Repetition Segmentation

Segmentation must be cheap. Because the repertoire is known and strongly periodic, we use phase detection on a single tracked coordinate rather than a general-purpose period predictor, in the manner of [18], [19].

The tracked scalar is one coordinate of one landmark, selected per exercise, so that vertical movements track Y and horizontal movements track X. Raw coordinates are smoothed by a trailing mean over a FIFO buffer of $N = 30$ frames,

$$\bar{p}_t = \frac{1}{N} \sum_{i=t-N+1}^{t} p_i, \tag{5}$$

and the current sample is compared against this level with a dead band Δ_{thresh}:

$$\mathrm{down}_t = \big[p_t > \bar{p}_t + \Delta_{\mathrm{thresh}}\big], \quad \mathrm{up}_t = \big[p_t < \bar{p}_t - \Delta_{\mathrm{thresh}}\big], \tag{6}$$

with $\Delta_{\mathrm{thresh}} = 0.1$ in normalised units. The dead band prevents state chattering when the signal sits near the mean, which is the dominant failure mode of a plain threshold. A repetition is registered on a completed up-to-down transition,

$$r \mathrel{+}= 1 \iff \mathrm{down}_t \wedge \mathrm{up}_{t-1}, \tag{7}$$

and for compound movements the transition must be confirmed by a second landmark (hip as primary and knee as secondary for a squat), which suppresses counts produced by isolated movement of one segment.

Per-frame cost is $O(1)$ amortised with $O(N)$ memory, which is what allows segmentation to run alongside inference inside the frame budget.

E. CONSTRAINT EVALUATION AND FEEDBACK

Movement-quality assessment can be posed as classification over known error labels, and solved with a learned classifier or an explicit constraint base. The learned route requires a labelled corpus per error class per exercise, which does not exist at the scale the repertoire demands. VIRTA therefore evaluates geometric constraints defined per exercise against biomechanically motivated ranges. The decision variable is a joint angle and the joint angle is what the feedback names, so interpretability (R3) is structural rather than retrofitted.

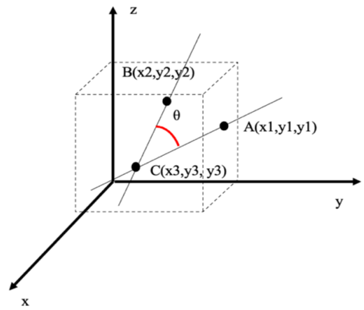

FIGURE 10. Angle θ at landmark C spanned by the segments to A and B, evaluated by Eq. (8).

1) Angle Computation

The primary feature is the angle at a joint spanned by two adjacent segments. Given landmarks $\mathbf{p}_a, \mathbf{p}_b, \mathbf{p}_c$ with the joint at $\mathbf{p}_b$ (Fig. 10), form $\mathbf{v}_1 = \mathbf{p}_a - \mathbf{p}_b$ and $\mathbf{v}_2 = \mathbf{p}_c - \mathbf{p}_b$; then

$$\theta = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\|\,\|\mathbf{v}_2\|}\right). \tag{8}$$

Two numerical guards are part of the implementation. The cosine is clipped to $[-1, 1]$ before the $\arccos$, because accumulated floating-point error in the dot product otherwise yields NaN at the fully extended and fully flexed limits—precisely the configurations the constraints test. A zero-length vector, which occurs when two landmarks coincide under tracking failure, returns 0° and exits early rather than dividing by zero.

2) Per-Exercise Constraint Sets

For each exercise a small set of landmark triples is designated as diagnostic. For the squat these are the symmetric hip, shoulder and knee joints (Fig. 11), which together capture flexion depth, trunk inclination and left–right symmetry. Each diagnostic angle is compared against a range representing safe execution; for the squat, knee flexion at the bottom position is expected within 70° to 100°, and departures on either side are distinct errors with distinct corrections. Constraints are not restricted to angles: normalised relative distances capture alignment requirements such as knee-over-toe displacement, $\tilde{d}_{\text{knee-toe}} < \delta$, and first differences of angles capture uncontrolled tempo.

Exercise-specific constraint sets were authored for the movements used in the evaluation, with joint-angle bounds derived from the reference executions and reviewed for biomechanical plausibility before deployment.

Aggregating over the repetition rather than reporting per frame is what makes the output usable: a transient violation

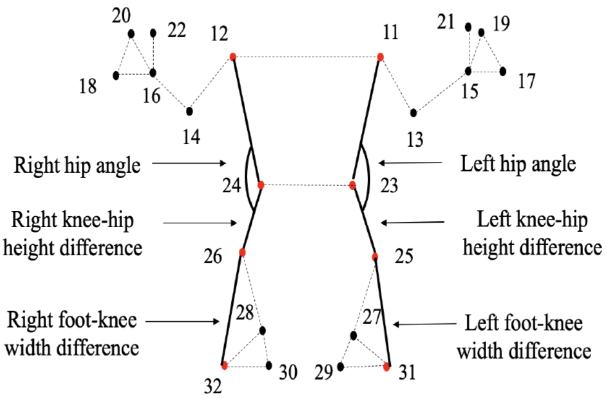

FIGURE 11. Landmarks used to validate squat posture, with landmark indices. Bilateral hip angles capture depth and trunk inclination; knee–hip height and foot–knee width differences capture asymmetry.

Algorithm 2 Repetition-level constraint evaluation

Require: frames $\mathbf{P}^{(t_0)}, \ldots, \mathbf{P}^{(t_1)}$ of one repetition; constraint set $\mathcal{K}_a$

$\mathcal{F} \leftarrow \emptyset$
for $t = t_0$ **to** t_1 **do**
  for each $\kappa = (\mathbf{p}_a, \mathbf{p}_b, \mathbf{p}_c, \theta_{\min}, \theta_{\max}, m^-, m^+) \in \mathcal{K}_a$ **do**
    $\theta \leftarrow$ Eq. (8) on κ at frame t
    if $\theta < \theta_{\min}$ **then**
      $\mathcal{F} \leftarrow \mathcal{F} \cup \{m^-\}$
    else if $\theta > \theta_{\max}$ **then**
      $\mathcal{F} \leftarrow \mathcal{F} \cup \{m^+\}$
    end if
  end for
end for
if $\mathcal{F} = \emptyset$ **then**
  return reinforcement message
end if
return highest-priority element of $\mathcal{F}$

at the turnaround of a movement is normal, and only a violation characterising the repetition is worth reporting. At most one message is emitted per repetition (R5), ordered so that safety-relevant violations pre-empt range-of-motion ones, and a message about a given constraint is suppressed for a cooldown interval after delivery so that the trainee has time to act before being told again. Repeated feedback for the same constraint is suppressed for 5 s after delivery.

3) Feedback Realisation and Delivery

Each violation maps to a short imperative utterance through a fixed template whose variant is selected by the sign of the deviation: a knee angle below $\theta_{\min}$ yields "go deeper on the next one", above $\theta_{\max}$ yields "that is deep enough, come up sooner", and a trunk inclination violation yields "keep your chest up". Template realisation is deliberate rather than expedient: the utterance inventory is finite and auditable, and output is limited to authored phrases, which is what allows a deployment to be reviewed by a coach or clinician before it reaches users.

Selected utterances are synthesised by a cloud neural text-to-speech service, Microsoft Azure Neural TTS [31], which returns audio together with a phoneme boundary stream. Viseme weights on the trainer's face are driven from that stream, so the correction is spoken by the demonstrating agent rather than played as a system sound. The smartphone measurement node and the VR presentation node exchange data over WebRTC on the local network—Wi-Fi 6 (802.11ax) at 5 GHz, static addressing on a private subnet—and only the feedback token crosses that link: exercise identifier, repetition index, constraint identifier, sign, magnitude, and the realised text. No frame and no landmark coordinate is transmitted off the smartphone, which is what R4 requires. The feedback text, which carries no identifying information, is the one payload that leaves the local network, going to the cloud speech service; the system is therefore local except for speech synthesis, and we do not claim otherwise.

TABLE 2. Deployment Configuration

Component	Specification
Reference capture	Rokoko Smartsuit Pro II; all 19 9-DoF IMUs active at 100 Hz; 5000 mA h hub (sensor count: see Section IV-B)
Measurement device	Google Pixel 7 Pro
Mobile SoC / OS	Google Tensor G2; Android 13
Camera	Main rear RGB camera; 12.5 MP effective (4080 × 3072, pixel-binned from 50 MP); 24 fps capture; 82° diagonal field of view (≈70° horizontal, ≈55° vertical); 24 mm-equivalent lens, $f/1.85$, 1/1.31-inch sensor
Pose inference input	640 × 480
Pose model	BlazePose Lite, 1.3 M parameters
Runtime	MediaPipe, GPU delegate; 8-bit post-training quantisation and pruning
Application	Flutter/Dart with embedded Unity, on Android
Backend	Firebase / Firestore (exercise library)
Presentation node	Meta Quest 2; Meta Quest OS, OpenXR runtime v50+
Avatar pipeline	84-bone humanoid rig, 72 blend shapes, 9 visemes
Speech synthesis	Microsoft Azure Neural TTS, voice `en-US-AndrewNeural`
Phone–headset link	WebRTC over Wi-Fi 6 (802.11ax), 5 GHz, static IP on a local subnet
Camera placement	Vertical, facing the trainee, whole body in frame (Section III-B)

VI. EXPERIMENTAL SETUP

A. APPARATUS

Table 2 lists the deployment configuration. The measurement node is a smartphone running the companion application, built in Flutter with the rendering subsystem embedded as a Unity view; camera frames reach the inference stage as GPU buffers. The presentation node renders the virtual trainer, whose avatars carry a humanoid rig, 72 facial blend shapes for expression and 9 visemes for lip synchronisation [32].

B. DATASETS AND PROTOCOLS

The evaluations below were carried out for the author's dissertation [6], Sections 6.1–6.5, of which this system is the subject; this article is their first journal presentation. Quantities written with $\pm$ are means and standard deviations, not confidence intervals.

Pose accuracy. A held-out test set of 2500 frames collected from 20 participants under varied illumination, background complexity and partial occlusion was scored by mean per-joint position error (MPJPE). Ground-truth body motion was obtained using inertial markers recorded concurrently with the smartphone stream. At the start of each take, the participant held the neutral pose and performed a deliberate arm-raise synchronisation gesture that was visible in both streams. The frame of maximum wrist elevation was used as the common temporal anchor, after which the 100 Hz inertial trajectory was linearly interpolated at the smartphone frame times. A single rigid transform estimated from the neutral pose registered the two coordinate systems: the hip midpoint defined the root, gravity defined the vertical axis and the shoulder line defined the mediolateral axis. MPJPE was computed after removing root translation while preserving the recorded metric scale; no scale fitting was applied.

Segmentation and error detection. 150 exercise sessions spanning resistance-training and rehabilitation movements. Segmentation is scored by temporal overlap with annotated repetition boundaries at $\mathrm{IoU} \geq 0.5$; error detection by precision and recall against annotated form violations. Repetition boundaries and biomechanical form violations were manually annotated from the recorded exercise sessions. The author performed the annotation frame by frame. A repetition began when the tracked trajectory left the exercise's neutral region and ended when it returned after completing the movement cycle. A form violation was labelled when the relevant joint angle or alignment remained outside the reference-derived permissible range during the movement phase to which that constraint applied; isolated transition-frame excursions were not labelled.

Deployment study. A four-week study with two cohorts: 20 amateur exercisers performing three supervised sessions per week, and 5 professional athletes using the system as supplementary feedback within an existing regimen. The amateur cohort was 20–35 years old and comprised 16 male and 4 female participants.

External comparison. Section VII-E summarises a published Kinect–MediaPipe comparison [33] as literature context rather than as a measurement of VIRTA.

C. ETHICAL CONSIDERATIONS

All study procedures complied with the ethical principles outlined in the Declaration of Helsinki. Written informed consent was obtained from all participants prior to participation. Participants were informed about the study procedures and potential virtual-reality-related side effects, including cybersickness, dizziness, headache, eye strain and temporary disorientation. Participants were instructed to report any discomfort during the experimental sessions and were free to pause or discontinue participation at any time.

The study involved non-invasive movement assessment using smartphone-based pose estimation and a virtual-reality headset used for presentation. The trainee wore no sensing instrumentation; all movement sensing was performed through the smartphone camera. No medical intervention or treatment was administered. Participant data were anonymised prior to analysis, and no personally identifiable information was included in the reported results. Camera imagery was processed on the participant's device and was neither stored nor transmitted.

No participant required discontinuation because of virtual-reality-related adverse effects.

TABLE 3. 3D Pose Accuracy Across Environmental Conditions

Environmental scenario	MPJPE (mm)
Controlled laboratory conditions	28.5 ± 4.3
Mobile environment, partial occlusion	36.8 ± 5.7
Mobile environment, low illumination	42.1 ± 6.2

VII. RESULTS

A. POSE ESTIMATION ACCURACY

Table 3 reports MPJPE across environmental conditions. Under controlled conditions the measurement tier achieves 28.5 mm. Partial occlusion, in which limbs are transiently outside the camera's view, raises the error to 36.8 mm, and low illumination raises it further to 42.1 mm—increases of 29 % and 48 % respectively over the controlled condition. Low light is therefore the harder of the two domestic conditions, and evening training, a common deployment case for home exercise, is where preprocessing or supplementary illumination would pay for itself first.

Mean per-frame inference time is (42 ± 8) ms, an inference-only rate of 23.8 fps, which meets the 20 fps floor of R2 for the inference stage.

B. EXERCISE CLASSIFICATION

The DTW nearest-neighbour classifier of Section V-D1 selects the constraint set applied to the whole set of repetitions, so its accuracy conditions every downstream result. It was evaluated over the two exercise classes used for downstream assessment (squat and downward-facing dog) on the 150 held-out session sequences. It correctly classified 141 sequences, for an overall accuracy of 94.0 %. The predicted label, rather than a user-selected label, was used to choose the constraint set for the segmentation and error-detection results below.

C. SEGMENTATION AND ERROR DETECTION

Table 4 reports segmentation and detection performance. Repetition segmentation reaches 94.7 % temporal-overlap accuracy on squats and 92.4 % on rehabilitation postures. Error detection reaches precision 0.91 and recall 0.89 on

TABLE 4. Segmentation and Error Detection Performance Over 150 Sessions

Metric	Squats	Rehabilitation
Segmentation accuracy (IoU ≥ 0.5)	94.7 %	92.4 %
Detection precision	0.91	0.88
Detection recall	0.89	0.85

TABLE 5. Measured Stage Timings and Perceptual Quality

Metric	Value
Pose inference, per frame	(42 ± 8) ms
Rendering and animation update	(58 ± 10) ms
Speech synthesis	(180 ± 25) ms
Network transfer (WebRTC, local)	12 ms to 18 ms
Speech naturalness, MOS (1–5)	4.3 ± 0.4 ($n = 15$)

squats, covering knee valgus, insufficient depth and forward trunk lean, and 0.88 and 0.85 on rehabilitation exercises—principally downward-facing dog—covering hip asymmetry, shoulder protraction and spinal misalignment. The corresponding F_1 scores are 0.90 and 0.87. Detection is thus slightly more precise than sensitive in both categories: the system misses more deviations than it invents, which is the error asymmetry to prefer when a false correction costs user trust.

D. LATENCY AND RESPONSIVENESS

Table 5 gives the measured stage timings. Pose inference costs (42 ± 8) ms per frame, the rendering and animation update (58 ± 10) ms, transfer of the feedback token from phone to headset over local WebRTC 12 ms to 18 ms, and speech synthesis (180 ± 25) ms—the last dominating the loop and being the one stage executed off the local network. Synthesised speech was rated 4.3 ± 0.4 for naturalness on a five-point scale by 15 raters.

The quantity a trainee experiences is the delay between the end of a repetition and the first audible phoneme of the correction for it. The stages that contribute after the repetition boundary are constraint evaluation, network transfer, speech synthesis and rendering; taken serially the measured ones sum to roughly 250 ms. That post-repetition figure should not be confused with the movement-to-correction delay, which is dominated by the structural term of Eq. (1).

Two properties of the loop follow from these figures and matter for the interaction. First, the response to a repetition is bounded below by the repetition itself: the structural term Δ_{seg} of Eq. (1) is a full repetition period, approximately 2 s for a squat, so VIRTA corrects the next repetition rather than the current one. Second, the 30-frame trailing mean of Eq. (5) has a group delay of $(N - 1)/2$ samples, about 0.6 s at the measured frame rate, which sets how sharply a repetition boundary can be located.

E. PUBLISHED KINECT–MEDIAPIPE COMPARISON

This subsection reports external evidence for context, not measurements of VIRTA. Cóias et al. [33] compared MediaPipe BlazePose with Kinect v2 on three upper-limb tasks performed by 15 stroke survivors. Table 6 gives their World Landmarks results. Coordinates were centred at the spine base and each trajectory standardised by its own mean and standard deviation, so the errors are dimensionless rather than MPJPE in millimetres, and only shared, successfully processed frames were included.

TABLE 6. Published MediaPipe World Landmarks Versus Kinect v2: Normalised Trajectory RMSE (Mean ± SD), Selected Arm Joints, From Cóias *et al.* [33], Table III

Joint	x	y	z
Left shoulder	0.67 ± 0.33	0.87 ± 0.38	1.18 ± 0.33
Left elbow	0.63 ± 0.17	0.65 ± 0.28	1.14 ± 0.32
Left wrist	0.61 ± 0.23	0.73 ± 0.24	1.19 ± 0.34
Right shoulder	0.66 ± 0.22	0.72 ± 0.28	1.31 ± 0.35
Right elbow	0.64 ± 0.20	0.55 ± 0.18	1.06 ± 0.39
Right wrist	0.75 ± 0.31	0.62 ± 0.19	1.20 ± 0.22

The pattern of interest is that depth-axis disagreement exceeds image-plane disagreement at every joint, by roughly a factor of two. This is the published counterpart of the design decision taken in Section V-B: VIRTA builds its constraints from joint angles and torso-normalised distances, which are dominated by the image-plane components, rather than from absolute depth. Kinect is the comparator in that study, not an independent optical ground truth, and the study does not speak to squat assessment or to the mobile model variant deployed here.

F. DEPLOYMENT STUDY

Over four weeks, amateur participants ($n = 20$) reduced their rate of biomechanical deviation events per session by 21.3 % between baseline and the final week. The outcome measure is the system's own detector: these are deviations as VIRTA counts them, and the study is single-arm and descriptive, so the figure documents engagement with the feedback rather than a demonstrated gain in movement quality. Section IX states what a study capable of the stronger claim would require.

Self-reported satisfaction averaged 4.4 ± 0.5 on a five-point scale, with participants citing the objective and non-judgemental character of automated assessment, and 35 % reported increased confidence in their form. In the professional cohort ($n = 5$), the system reached precision 0.92 and recall 0.88 against expert assessment of technique deviations, and cohort members confirmed that flagged errors aligned with what they would expect from their coaching staff. Table 7 collects the outcomes.

VIII. DISCUSSION

A. WHAT THE TWO-TIER SPLIT BUYS

The central design decision is to instrument the reference and not the trainee. It buys three things at once. The demonstrated target and the assessment standard are the same artefact, so a trainee is never judged against a criterion different from the movement they were shown. The instrumentation cost is

TABLE 7. Four-Week Deployment Outcomes

Outcome	Value
Reduction in system-detected deviation events ($n = 20$)	21.3 %
Satisfaction, 1–5 ($n = 20$)	4.4 ± 0.5
Reported increased confidence	35 %
Precision vs. expert ($n = 5$)	0.92
Recall vs. expert ($n = 5$)	0.88

paid once, offline, by the instructor, so it does not scale with the number of users—which is what makes a 180-movement library affordable outside a laboratory. And the online tier is free to be a commodity phone, because it is not required to establish ground truth on its own, only to measure a trainee against a standard that was established elsewhere.

The results are consistent with that division of labour. The pose tier degrades gracefully rather than failing: 28.5 mm controlled, 42.1 mm in the worst domestic condition tested, over a range of conditions a living room actually produces. Detection precision of 0.91 with recall of 0.89 on squats is achieved with no learned error classifier and no labelled error corpus—the constraint bank is authored, so adding an exercise costs a capture take and a handful of angle bounds rather than a labelling campaign. For a system whose purpose is deployability, that scaling property matters more than the last point of accuracy.

B. INTERPRETABILITY AS AN ENGINEERING PROPERTY

Because the decision variable is a joint angle and the joint angle is what the utterance names, the path from sensor to spoken sentence is inspectable at every step: a coach can read the constraint set for an exercise, a clinician can audit the finite utterance inventory before deployment, and a failure can be traced to a specific angle on a specific repetition. This is a direct consequence of choosing an explicit constraint base over a learned classifier, and it is the property that makes the system reviewable by the professionals who would deploy it. The corresponding cost is coverage: only movements with an authored constraint set can be assessed at all.

C. FEEDBACK TIMING

A repetition-level correction necessarily describes a movement that has already ended, and the measured stage timings put the earliest possible delivery well inside the following repetition. The design consequence, realised in R5 and in the cooldown of Section V-E, is that the interaction is framed around the next repetition: the utterance inventory is written in that tense (“go deeper on the next one”), and at most one instruction is issued per repetition so that a trainee under exertion is not tracking a queue of stale corrections. Speech was the right channel for this reason rather than for an aesthetic one—it is the only channel that reaches a trainee who is mid-movement and cannot look at a display.

D. PRIVACY AS A DEPLOYMENT PRECONDITION

Camera imagery of a person exercising at home is among the more sensitive data a consumer system can collect, and the willingness to install such a system turns on what leaves the room. In VIRTA the frames are consumed as GPU buffers and discarded, the landmark coordinates never leave the process, and the only payload crossing a network boundary is a short feedback token and the authored sentence to be synthesised. That property is architectural rather than a policy setting, which is what makes it a claim a deployer can verify.

IX. LIMITATIONS

The results of Section VII describe one implementation measured under the protocols of Section VI-B. This section states, in one place, what those results do and do not establish.

A. EVALUATION DESIGN

The pose accuracy figures are descriptive rather than comparative. They were not obtained on a public benchmark, so they do not license a ranking against other pose models, and the reported dispersions are standard deviations rather than confidence intervals. The ground-truth instrument against which the error was scored was an inertial-marker system, and the test set comprised 20 participants. The reported values use event-based temporal synchronisation, neutral-pose rigid registration and root-translation alignment without scale fitting. They should therefore be interpreted as root-aligned errors at the retained metric scale, not as raw camera-coordinate errors or Procrustes-aligned benchmark scores. No controlled comparison against an alternative pose backbone was run, so the choice of BlazePose is justified by its operating point rather than by a measured advantage over a competitor on this task.

The four-week deployment is single-arm. It has no control condition and no independent movement-quality outcome: the 21.3 % reduction is a reduction in events counted by the system being evaluated, which cannot separate improved technique from adaptation to the detector. It therefore supports no claim of motor learning, of injury prevention, or of a benefit attributable to immersion or to embodiment specifically. The five-athlete cohort is a case series. Participants were briefed on the possible virtual-reality side effects and monitored for discomfort throughout, and none discontinued for that reason, but no standardised presence, workload, cybersickness or retention instrument was administered, so the absence of reported adverse effects is not a measured tolerability result. Establishing that the embodied interface helps would require a controlled comparison holding the reference motions, the sensing and the correction content constant while varying only presentation and feedback availability, scored by blinded raters.

Repetition boundaries and biomechanical form violations were manually annotated by the author using the criteria in Section VI-B. Because a single annotator was used, interrater agreement was not measured; the reported precision and recall remain conditional on a single-rater reference standard.

B. SENSING AND ASSESSMENT

A single camera facing the trainee is subject to self-occlusion, depth ambiguity and poor coverage of floor-based postures; the published cross-modal comparison of Section VII-E shows depth-axis disagreement roughly twice the image-plane disagreement, which is why the constraint bank avoids absolute depth. A head-mounted display may itself occlude landmarks the pose pipeline uses, and no headset-worn versus headset-free benchmark was run.

Three properties of the evaluated implementation bound its reliability and are the first candidates for revision. Constraint evaluation is not gated by the per-landmark visibility score, so a transiently mistracked joint can be assessed as though it were observed. The repetition-level rule takes the union of frame-level violations without requiring persistence, so a single noisy frame can flag an otherwise correct repetition. And the segmenter's Δ_{thresh} is a fixed absolute dead band against a trailing mean of the signal itself, so a very slow repetition can drag the reference level towards the current sample and suppress the transition, while a set performed with progressively decreasing range of motion—the normal signature of fatigue—produces a shrinking excursion against a constant threshold. Each of these changes would need held-out evaluation of false alarms and missed deviations before deployment, not merely implementation.

The joint-angle bounds are fixed and were not validated across body proportions, sex, age or legitimate variations in exercise technique, so a trainee whose correct execution falls outside an authored range will be corrected wrongly. Template realisation guarantees that the system says only authored sentences; it does not guarantee that the sentence is biomechanically right, since an erroneous landmark or an inappropriate threshold produces a well-formed but incorrect correction. The fixed 5 s feedback cooldown was chosen to prevent repeated prompts during adjacent repetitions; it was not independently tuned or compared against other durations.

The classification experiment covers only the two exercise classes used for downstream assessment. Its 94.0 % accuracy therefore does not establish recognition across the complete 180-movement reference library, and a misclassification still means the trainee is assessed against another exercise's bounds. The DTW classifier also compares a flattened representation, with no comparison against a multivariate alignment that preserves landmark structure.

C. COVERAGE AND SCOPE

The library contains 180 captured movements, which is not the same as 180 assessable exercises: exercise-specific constraint sets were authored for the movements used in the evaluation. The documented examples are squats and downward-facing dog. The reference motions define a traceable but not universal assessment standard; a reusable trainer should declare which technique variant it encodes and which constraints it applies.

X. CONCLUSION

This article presented VIRTA, an embodied exercise trainer that connects inertially captured demonstrations with mobile pose sensing, geometric assessment and spoken repetition-level feedback in virtual reality. Its organising idea is to instrument the reference rather than the trainee: the inertial suit is worn once, offline, by an instructor, and the resulting clips serve both as the animation the trainee watches and as the templates against which the trainee is measured. During a session the trainee wears no sensing instrumentation, a single smartphone performs all sensing and inference on device, and no image or landmark data leaves the phone.

The measured behaviour of the deployed configuration is a mean per-joint position error of 28.5 mm under controlled conditions, rising to 42.1 mm in low light; 42 ms mean inference time per frame; 94.0 % exercise-classification accuracy across the two assessed movements; 94.7 % repetition-segmentation accuracy and detection precision and recall of 0.91 and 0.89 on squats; and, in a four-week single-arm deployment with 20 amateur exercisers, a 21.3 % reduction in system-detected deviation events at a satisfaction rating of 4.4 ± 0.5. The scope of these results is set out in Section IX.

The work suggests three next steps in order of leverage. Gating constraint evaluation by landmark visibility and requiring violations to persist across a validated fraction of the repetition should reduce false corrections, which are the failures that cost user trust most directly. Instrumenting feedback arrival against repetition boundaries with an event trace would replace stage-level timings with the quantity the interaction actually depends on. And a controlled comparison against matched non-embodied presentation, scored by independent raters, is what would establish whether the embodied trainer improves guidance beyond the sensing and animation components it is built from.

ACKNOWLEDGMENT

The author thanks the motion-capture demonstrators who performed the reference library, the coaches who reviewed the constraint thresholds, and the participants of the deployment study.

REFERENCES

[1] M. Fieraru, M. Zanfir, S. C. Pirlea, V. Olaru, and C. Sminchisescu, "AIFit: Automatic 3D human-interpretable feedback models for fitness training," in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021.

[2] F. Hülsmann, J. P. Göpfert, B. Hammer, S. Kopp, and M. Botsch, "Classification of motor errors to provide real-time feedback for sports coaching in virtual reality — a case study in squats and Tai Chi pushes," Computers & Graphics, vol. 76, pp. 47–59, 2018.

[3] I. H. López-Nava and A. Muñoz-Meléndez, "Wearable inertial sensors for human motion analysis: A review," IEEE Sensors Journal, vol. 16, no. 22, pp. 7821–7834, 2016.

[4] S. Ciklacandir, S. Ozkan, and Y. Isler, "A comparison of the performances of video-based and IMU sensor-based motion capture systems on joint angles," in Innovations in Intelligent Systems and Applications Conference (ASYU), 2022, pp. 1–5.

[5] M. Aslanyan, “Development of intelligent workout environment for VR devices,” in IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 1–5.
[6] ——, “Development of an intelligent environment for the automated analysis of human motor behavior,” Candidate of Technical Sciences dissertation, Institute for Informatics and Automation Problems, National Academy of Sciences of the Republic of Armenia, Yerevan, Armenia, 2026.
[7] ——, “On mobile pose estimation and action recognition design and implementation,” Pattern Recognition and Image Analysis, vol. 34, pp. 126–136, 2024.
[8] ——, “Comparative evaluation of inertial measurement unit versus vision based motion capture,” Pattern Recognition and Image Analysis, vol. 35, no. 4, pp. 953–964, 2025.
[9] S. Mihcin, H. Kose, S. Cizmeciogullari, S. Ciklacandir, M. Kocak, A. Tosun, and A. Akan, “Investigation of wearable motion capture system towards biomechanical modelling,” in IEEE International Symposium on Medical Measurements and Applications (MeMeA), 2019.
[10] S. Scataglini, E. Abts, C. Van Bocxlaer, M. Van den Bussche, S. Meletani, and S. Truijen, “Accuracy, validity, and reliability of markerless camera-based 3D motion capture systems versus marker-based 3D motion capture systems in gait analysis: A systematic review and meta-analysis,” Sensors, vol. 24, no. 11, p. 3686, 2024.
[11] S. Johnson and M. Everingham, “Clustered pose and nonlinear appearance models for human pose estimation,” in British Machine Vision Conference (BMVC), 2010.
[12] M. Andriluka, L. Pishchulin, P. Gehler, and B. Schiele, “2D human pose estimation: New benchmark and state of the art analysis,” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014.
[13] L. Sigal, A. O. Balan, and M. J. Black, “HumanEva: Synchronized video and motion capture dataset and baseline algorithm for evaluation of articulated human motion,” International Journal of Computer Vision, vol. 87, no. 1–2, pp. 4–27, 2010.
[14] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, “Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 36, no. 7, pp. 1325–1339, 2014.
[15] A. Newell, K. Yang, and J. Deng, “Stacked hourglass networks for human pose estimation,” in European Conference on Computer Vision (ECCV), 2016, pp. 483–499.
[16] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, “BlazePose: On-device real-time body pose tracking,” in CVPR Workshop on Computer Vision for Augmented and Virtual Reality, 2020.
[17] V. Bazarevsky, Y. Kartynnik, A. Vakunov, K. Raveendran, and M. Grundmann, “BlazeFace: Sub-millisecond neural face detection on mobile GPUs,” in CVPR Workshop on Computer Vision for Augmented and Virtual Reality, 2019.
[18] I. Pernek, K. A. Hummel, and P. Kokol, “Exercise repetition detection for resistance training based on smartphones,” Personal and Ubiquitous Computing, vol. 17, no. 4, pp. 771–782, 2013.
[19] D. Morris, T. S. Saponas, A. Guillory, and I. Kelner, “RecoFit: Using a wearable sensor to find, recognize, and count repetitive exercises,” in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ’14). Toronto, ON, Canada: ACM, 2014, pp. 3225–3234.
[20] S. Sempena, N. U. Maulidevi, and P. R. Aryan, “Human action recognition using dynamic time warping,” in International Conference on Electrical Engineering and Informatics (ICEEI), 2011.
[21] D. J. Berndt and J. Clifford, “Using dynamic time warping to find patterns in time series,” in Proceedings of the AAAI-94 Workshop on Knowledge Discovery in Databases (KDD-94), ser. AAAI Technical Report WS-94-03. Seattle, WA, USA: AAAI Press, 1994, pp. 359–370, https://aaai.org/papers/359-ws94-03-031/.
[22] P. T. Chua, R. Crivella, B. Daly, N. Hu, R. Schaaf, D. Ventura, T. Camill, J. Hodgins, and R. Pausch, “Training for physical tasks in virtual environments: Tai Chi,” in IEEE Virtual Reality, 2003, pp. 87–94.
[23] F. Tian, S. Ni, X. Zhang, F. Chen, Q. Zhu, C. Xu, and Y. Li, “Enhancing Tai Chi training system: Towards group-based and hyper-realistic training experiences,” IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 5, 2024.
[24] J. Liu, N. Saquib, Z. Chen, R. H. Kazi, L.-Y. Wei, H. Fu, and C.-L. Tai, “PoseCoach: A customizable analysis and visualization system for video-based running coaching,” IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 7, pp. 3180–3195, 2024.
[25] Y. Wu, L. Yu, J. Xu, D. Deng, J. Wang, X. Xie, H. Zhang, and Y. Wu, “AR-enhanced workouts: Exploring visual cues for at-home workout videos in AR environment,” in Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST ’23). New York, NY, USA: ACM, 2023.
[26] X. Yu, B. Lee, and M. Sedlmair, “Design space of visual feedforward and corrective feedback in XR-based motion guidance systems,” in Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). ACM, 2024, pp. 1–15.
[27] S. Eom, T. Hu, W. Xu, L. Zou, E. Escobar, G. Streisfeld, A. Mall, B. Granger, and M. Gorlatova, “Rhythms of recovery: Patient-centered virtual reality exergame for physical rehabilitation in the intensive care unit,” IEEE Transactions on Visualization and Computer Graphics, 2026.
[28] H. Cheng, S. J. H. Ong, S. Cai, A. T. Y. Koh, F. Ouyang, and E. T. Khoo, “EgoPoseVR: Spatiotemporal multi-modal reasoning for egocentric full-body pose in virtual reality,” arXiv preprint, 2026.
[29] J. Stenum, C. Rossi, and R. T. Roemmich, “Two-dimensional video-based analysis of human gait using pose estimation,” PLOS Computational Biology, vol. 17, no. 4, p. e1008935, 2021.
[30] F. Hülsmann, S. Kopp, and M. Botsch, “Automatic error analysis of human motor performance for interactive coaching in virtual reality,” arXiv preprint, 2017.
[31] Microsoft, “Text to speech — azure AI speech documentation,” https://learn.microsoft.com/azure/ai-services/speech-service/text-to-speech, 2025, accessed: 2026-09-06.
[32] S. Altundas and E. Karaarslan, “Cross-platform and personalized avatars in the metaverse: Ready Player Me case,” in Digital Twin Driven Intelligent Systems and Emerging Metaverse. Springer Nature Singapore, 2023, pp. 317–330.
[33] A. R. Cóias, M. H. Lee, A. Bernardino, and A. Smailagic, “Skeleton tracking solutions for a low-cost stroke rehabilitation support system,” in International Conference on Rehabilitation Robotics (ICORR), 2023, pp. 1–6.

MINAS ASLANYAN is with the Discrete Mathematics Department, Institute for Informatics and Automation Problems, National Academy of Sciences of the Republic of Armenia, Yerevan, Armenia. His research focuses on human pose estimation, mobile and edge inference, virtual and augmented reality, and interactive systems for movement guidance. He develops deployable training and rehabilitation tools that combine inertial motion capture, monocular vision, geometric analysis and embodied feedback on commodity hardware. His broader interests include interpretable biomechanical assessment, privacy-preserving on-device processing, avatar animation and low-latency multimodal interaction. He has contributed to research on mobile pose estimation, action recognition and the practical comparison of vision-based and inertial motion-capture systems. He is particularly interested in translating computer-vision methods into accessible systems that operate outside specialist laboratories. His ORCID is 0009-0008-0094-3742.